# Confidence versus MC Dropout Mutual Information for Selective Pneumonia Classification under Gaussian Blur

**DONG YUAN**
MD6329 Deep Learning for Healthcare AI

## Abstract

A classifier can retain useful probability rankings while making poor decisions at a fixed threshold. We investigate this distinction in PneumoniaMNIST and compare confidence with Monte Carlo (MC) Dropout mutual information (MI) for selective classification. We train a compact convolutional network, a blur-augmented variant, and a classifier with dropout across three seeds. Models are evaluated on 624 official test images under clean inputs and three Gaussian blur levels. At the strongest blur level, baseline AUROC remains 0.8504, but specificity falls to zero at the fixed 0.5 threshold. Blur augmentation does not prevent near-universal pneumonia predictions. For the same MC predictions and a review budget of 124 images, low-confidence selection consistently retains fewer errors than high-MI selection. Their paired risk difference ranges from 1.27 to 4.00 percentage points across conditions. These findings favor a simple confidence reference in this restricted experiment, without establishing clinical safety or general superiority over other uncertainty methods.

## 1 Problem Formulation

For a healthcare image classifier, a high ranking metric is insufficient if its operating threshold produces excessive false positives. A review mechanism may remove unreliable predictions from automatic processing, but its benefit depends on whether the selection score actually identifies errors. Our objective is to study these relationships under a controlled change in image appearance rather than develop a clinically deployable diagnostic system.

We ask three questions. First, how do Gaussian blur and a fixed classification threshold affect ranking performance and class-specific errors? Second, does targeted blur augmentation reduce this degradation? Third, holding MC predictions fixed, does MI select errors more effectively than confidence in the mean predictive probability? The third question is the primary selective-classification comparison.

This is an empirical evaluation contribution: a reproducible comparison with paired selection decisions, independent training runs, and explicit failure analysis. We do not propose a new architecture or uncertainty estimator. Generative AI helped formulate the questions, develop code, and organize analysis; its outputs required execution checks and human review.

## 2 Related Work

Medical image learning motivates the application, including the image-based diagnosis study of Kermany et al. [1]. We use the standardized lightweight benchmark introduced by Yang et al. [2], rather than claim equivalence to the original full-resolution clinical task.

MD6329 course project. Research benchmark study.

Confidence is an important reference for error detection: Hendrycks and Gimpel [3] study maximum softmax probability as a simple baseline. Selective classification formalizes the trade-off between coverage and risk [4]. Our retrospective fixed-budget analysis follows this framing; it does not inherit the guarantees of a separately calibrated selective classifier.

Dropout was introduced as regularization [5]. Gal and Ghahramani provide an approximate Bayesian interpretation that motivates stochastic test-time predictions [6]. MI-based acquisition is also used in Bayesian active learning [7]. Selecting points informative for learning, however, is a different objective from identifying current classification errors; this distinction motivates our comparison rather than proving its outcome.

Controlled corruption benchmarks [8] and evaluations of uncertainty under dataset shift [9] motivate checking scores under degraded inputs. Deep ensembles offer another uncertainty approach [10], but are not implemented here. Precision-recall evaluation complements ROC analysis for imbalanced data [11]; we report average precision (AP), not an unspecified numerical area under a plotted PR curve.

Our design also connects directly to MD6329 Lecture 2, Medical Image Classification [13]. Slides 19 and 57 explain local convolutional feature extraction and the stacking of convolution, ReLU, pooling and classification layers; our compact CNN implements these operations with an additional global average pooling step. Slides 84 and 92 distinguish confusion-matrix measures from ROC analysis and discuss threshold trade-offs. Our blur experiment makes this distinction concrete: substantial AUROC can coexist with zero specificity at a fixed threshold. The cited locations are verified slide numbers in the course PDF, not unverified video timestamps.

## 3 Data

PneumoniaMNIST contains 5,856 pediatric chest radiographs represented as small grayscale images for normal-versus-pneumonia classification [1,2]. We preserve the official split: 4,708 training, 524 validation, and 624 test images. Test labels comprise 234 normal and 390 pneumonia cases. Images are represented as float32 tensors of shape 1 by 28 by 28 and divided by 255; labels are 0 for normal and 1 for pneumonia. We make no new patient-level independence claim beyond the supplied split.

The test pneumonia prevalence is 62.5%, whereas validation prevalence is approximately 74.2%. AP and accuracy should therefore not be interpreted as directly comparable across these sets without considering composition. An always-pneumonia test classifier achieves 62.5% accuracy, 100% sensitivity and zero specificity.

Gaussian blur uses a fixed 9 by 9 kernel with sigma 0.5, 1.0, or 1.5; the clean condition bypasses filtering. The same source samples and labels are used across conditions. These perturbations are controlled stress tests, not measurements of real hospital acquisition shifts. At 28 by 28 resolution, smoothing may remove substantial task information.

## 4 Approach

### 4.1 Models and training

SmallCNN consists of two 3 by 3 convolutions with padding 1 and 16 then 32 output channels. Each is followed by ReLU and 2 by 2 max pooling. Adaptive average pooling produces a 32-dimensional vector for a two-output linear classifier. The model has 4,866 trainable parameters. DropoutCNN adds dropout with probability 0.3 between flattening and the classifier, adding no trainable parameters.

The project builds conceptually on my Assignment 1 CNN exercise [14], which studied FashionMNIST with convolution, pooling and fully connected layers. Here, padded convolutions and global average pooling support a new two-class pneumonia task and blur and uncertainty experiments. Assignment 2 instead studied a synthetic clinical-text Transformer and PyVene interventions; it is not the method extended in this project. These connections distinguish prior coursework from the new study.

All models are trained from scratch with cross-entropy loss and Adam, learning rate 0.001, batch size 64, and a fixed ten-epoch budget. Training seeds are 42, 43 and 44. The checkpoint with lowest

clean validation loss is selected independently for each run. The blur-augmented model retains the clean input with probability 0.5; otherwise it applies Gaussian blur with sigma sampled uniformly from 0.3 to 1.0. Other training inputs remain clean. Sigma 1.5 lies outside this augmentation range.

The ten-epoch budget controls training effort but does not demonstrate convergence. Preliminary curves continued improving near epoch ten. Consequently, conclusions concern these training settings and may change with longer training or stronger architectures.

### 4.2 Inference and selection

Deterministic inference disables dropout. MC inference enables only the dropout module, disables gradient computation and performs T=30 stochastic forward passes. Let p_t be the pneumonia probability from the softmax of pass t and q be their mean. Probabilities, rather than logits, are averaged. Classification uses probability at least 0.5 for pneumonia.

MC confidence is max(q, 1-q). Binary entropy is H(p)=-p log(p)-(1-p)log(1-p). MI is H(q) minus the mean of H(p_t). Entropy calculations clamp probabilities away from zero and one. We compare ordinary deterministic confidence, MC mean confidence, and MC MI. The primary comparison uses the same MC mean labels for confidence and MI, changing only the review ranking.

Low-confidence or high-MI samples are removed from automatic processing. At target review fraction r, floor(Nr) samples are reviewed. For N=624 and r=0.2, 124 are reviewed and 500 retained: coverage is 0.801282. Risk is the error count divided by retained count. Ties use a fixed label-independent random order. A uniformly random retained subset has expected risk equal to the full-set error rate. We do not assume that human review is correct or estimate final combined-system accuracy.

### 4.3 Evaluation protocol

Validation experiments informed the settings; the protocol was frozen before final test inference. This is not preregistration before the exploratory study. For each run and condition we report AUROC, AP, accuracy, sensitivity and specificity. We compute the primary MI-minus-confidence risk difference within seed, then summarize those differences. Means and sample standard deviations describe three runs on the same test set; they are neither population confidence intervals nor evidence of statistical significance. Checkpoint restoration, file reloads and hashes provide consistency checks, not independent validation of all implementation details.

## 5 Experiments

### 5.1 Classification under blur

Clean baseline AUROC is $0.9132 \pm 0.0128$, with accuracy $0.7997 \pm 0.0262$ and specificity $0.5342 \pm 0.0864$. At sigma 1.5, AUROC remains $0.8504 \pm 0.0364$, while specificity becomes zero and accuracy becomes 0.6250 in every run. Every baseline prediction is then pneumonia. Ranking and fixed-threshold performance therefore tell different stories.

Blur augmentation does not resolve the threshold behavior. At sigma 1.5 its specificity is $0.0014 \pm 0.0025$: only one of three runs correctly classifies a single normal image. MC averaging does not consistently improve classification. For the dropout network, clean deterministic and MC AUROCs are 0.9101 and 0.9079; under strongest blur they are 0.8414 and 0.8338. These observations do not establish improved or degraded probability calibration, which was not directly evaluated.

### 5.2 Confidence and mutual information

At the same retained count, confidence leaves fewer errors than MI for every seed and condition. Under strongest blur this corresponds to 20 fewer errors on average among 500 retained images. Both average risks are below their matched theoretical random reference of 0.3745, but the remaining errors are substantial.

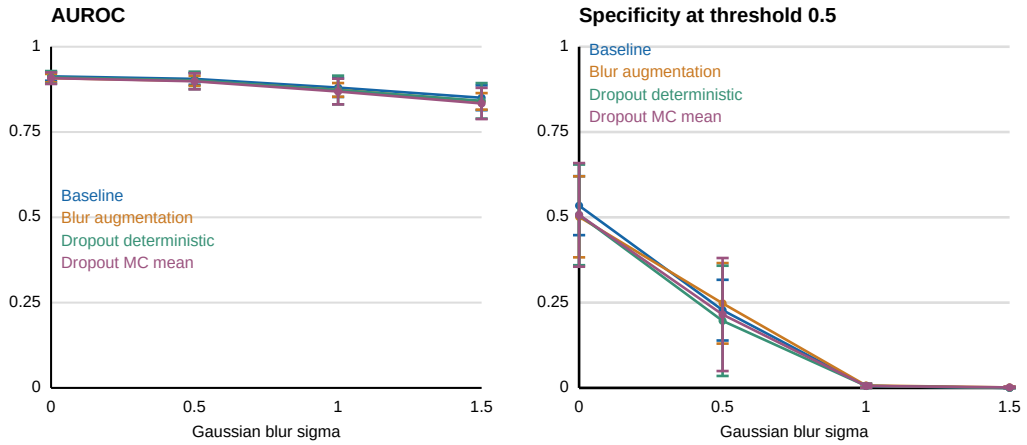


Figure 1: Test classification under controlled Gaussian blur. Error bars show sample SD across training seeds, not confidence intervals.

| $\sigma$ | Confidence risk | MI risk | Paired difference (pp) |
|---|---|---|---|
| 0 | $0.1580 \pm 0.0465$ | $0.1853 \pm 0.0401$ | $2.73 \pm 1.10$ |
| 0.5 | $0.2240 \pm 0.0408$ | $0.2367 \pm 0.0395$ | $1.27 \pm 0.58$ |
| 1.0 | $0.2607 \pm 0.0202$ | $0.2820 \pm 0.0216$ | $2.13 \pm 0.50$ |
| 1.5 | $0.2760 \pm 0.0236$ | $0.3160 \pm 0.0160$ | $4.00 \pm 0.87$ |

Table 1: Target 20% review. Mean $\pm$ sample SD across three seeds; differences computed within seed first.

### 5.3 Interpretation and limitations

The paired comparison controls prediction labels, so its difference reflects selection rather than changes in classification. A plausible explanation is that disagreement among stochastic classifier outputs need not align with actual error ranking. We have not directly established the mechanism: dropout only affects the final feature vector, and this limited approximation may constrain the result.

The study uses one downsampled dataset, one compact architecture family, a short fixed training budget, three seeds and one corruption type. We did not vary dropout placement, tune a deployed threshold, evaluate external hospitals or model real reviewers. Fixed-budget ranking on a completed batch does not guarantee the same coverage on future cases. Low retained risk can also reflect changes in the retained class composition, so class-specific errors must remain visible. AI-assisted coding accelerated iteration but also made explicit checks of label handling, probability averaging and denominators essential. The supplementary implementation specifies the repeated training and evaluation logic. The accompanying LecTrace presentation [12] uses the supplied seed42 model to illustrate selected computations on validation examples.

## 6 Conclusion

Controlled blur exposes a large mismatch between probability ranking and classification at a fixed threshold. Targeted augmentation and MC averaging do not eliminate near-universal pneumonia predictions in the strongest condition. For the same MC predictions, mean-probability confidence selects errors more effectively than MI at the tested review budget in all three runs. The practical lesson is to benchmark uncertainty scores against simple confidence using the actual selection objective. Future work should examine longer training, richer uncertainty approximations, operating-point selection on validation data and independent datasets, without treating this benchmark as evidence of clinical readiness.

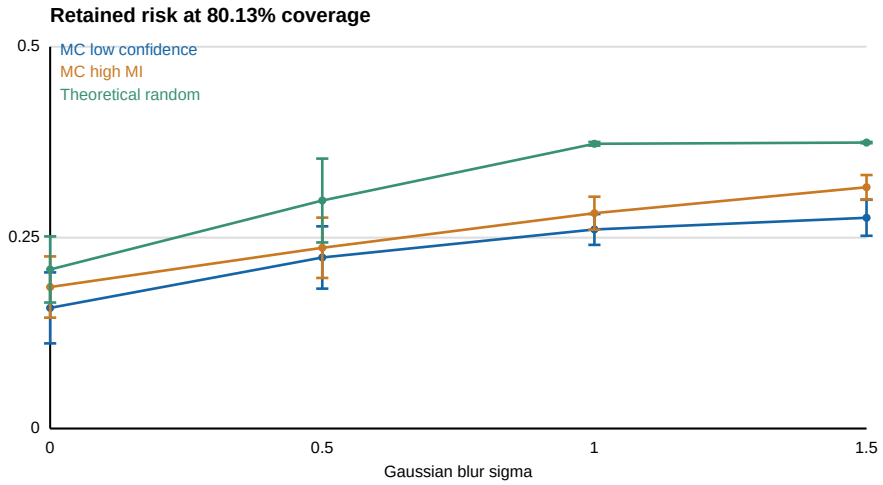


Figure 2: Risk after target 20 percent review. All B and C comparisons share MC labels; error bars are sample SD across seeds.

## References

[1] Kermany DS et al. Identifying Medical Diagnoses and Treatable Diseases by Image-Based Deep Learning. Cell, 172, 1122–1131.e9, 2018. https://doi.org/10.1016/j.cell.2018.02.010

[2] Yang J et al. MedMNIST v2 A large-scale lightweight benchmark for 2D and 3D biomedical image classification. Scientific Data, 10, 41, 2023. https://doi.org/10.1038/s41597-022-01721-8

[3] Hendrycks D and Gimpel K. A Baseline for Detecting Misclassified and Out-of-Distribution Examples in Neural Networks. ICLR, 2017. https://arxiv.org/abs/1610.02136

[4] Geifman Y and El-Yaniv R. Selective Classification for Deep Neural Networks. NeurIPS, 2017. https://papers.neurips.cc/paper_files/paper/2017/hash/4a8423d5e91fda00bb7e46540e2b0cf1-Abstract.html

[5] Srivastava N et al. Dropout A Simple Way to Prevent Neural Networks from Overfitting. JMLR, 15, 1929–1958, 2014. https://jmlr.org/papers/v15/srivastava14a.html

[6] Gal Y and Ghahramani Z. Dropout as a Bayesian Approximation Representing Model Uncertainty in Deep Learning. ICML, 2016. https://proceedings.mlr.press/v48/gal16.html

[7] Gal Y, Islam R and Ghahramani Z. Deep Bayesian Active Learning with Image Data. ICML, 2017. https://proceedings.mlr.press/v70/gal17a.html

[8] Hendrycks D and Dietterich T. Benchmarking Neural Network Robustness to Common Corruptions and Perturbations. ICLR, 2019. https://arxiv.org/abs/1903.12261

[9] Ovadia Y et al. Can you trust your model's uncertainty Evaluating predictive uncertainty under dataset shift. NeurIPS, 2019. https://papers.neurips.cc/paper_files/paper/2019/hash/8558cb408c1d76621371888657d2eb1d-Abstract.html

[10] Lakshminarayanan B, Pritzel A and Blundell C. Simple and Scalable Predictive Uncertainty Estimation using Deep Ensembles. NeurIPS, 2017. https://papers.neurips.cc/paper_files/paper/2017/hash/9ef2ed4b7fd2c810847ffa5fa85bce38-Abstract.html

[11] Saito T and Rehmsmeier M. The Precision-Recall Plot Is More Informative than the ROC Plot When Evaluating Binary Classifiers on Imbalanced Datasets. PLOS ONE, 10, e0118432, 2015. https://doi.org/10.1371/journal.pone.0118432

[12] PraiseGod D. LecTrace demo. Software and execution explanation examples. https://github.com/praisegee/lectrace-demo (accessed October 22, 2026).

[13] Tan CW. MD6329 Lecture 2 Medical Image Classification, slides 19, 57, 84 and 92. Course slides adapted from Serena Yeung, Stanford BIODS 220; relevant convolution material credits CS231n. NTULearn, accessed October 22, 2026. https://ntulearn.ntu.edu.sg/ultra/courses/_2704689_1/file/_6061108_1?courseId=_2704689_1

[14] DONG YUAN. MD6329 Assignment 1, Part 2 CNN Analysis using PyTorch, pp. 4–8. Submitted September 12, 2026. Unpublished course assignment.